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# DATABASE MANAGEMENT SYSTEMS
# [ENCS204]

**Bachelor of Technology**
**in**
**Computer Science and Engineering**

**Submitted by**
**[LAKSHITA KALRA](2401010117)**

**Course Teacher:**
**MISS MANSI KAJAL**

**School of Engineering and Technology**

**K.R.MANGALAM UNIVERSITY**
**Sohna,Haryana 122103,India**
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# LAB ASSIGNMENT- 3

## AIM:

To understand COMMIT and ROLLBACK in transaction control.To use ORDER BY and GROUP BY clauses. To understand DCL commands (GRANT, REVOKE) and SAVEPOINT.To perform JOIN operations in SQL.

## DESCRIPTION:

In this experiment, I created multiple tables and inserted records.
I applied UPDATE operations and used GROUP BY and ORDER BY clauses.I performed different types of JOIN operations.I also used transaction control commands like COMMIT, ROLLBACK, SAVEPOINT.
Finally, I implemented DCL commands such as GRANT and REVOKE.

## PROCEDURE:

## STEP 1: Create New Database

```
CREATE DATABASE DBMS_Lakshita_Lab3;
USE DBMS_Lakshita_Lab3;
```

| | | | | |
|---|---|---|---|---|
| 1 | 13:35:48 | CREATE DATABASE DBMS_Lakshita_Lab3 | 1 row(s) affected | 0.010 sec |
| 2 | 13:35:48 | USE DBMS_Lakshita_Lab3 | 0 row(s) affected | 0.00036 sec |

## DDL PART (TABLE CREATION)

## 1 Create TEACHER Table

```
CREATE TABLE TEACHER(
Name VARCHAR(50),
DeptNo INT,
Date_of_Joining DATE,
DeptName VARCHAR(50),
```

```
Location VARCHAR(50),
Salary INT
);
```

| | | | | |
|---|---|---|---|---|
| 3 | 13:37:24 | CREATE TABLE TEACHER( Name VARCHAR(50)... | 0 row(s) affected | 0.028 sec |

## 2 Create SALES Table

```
CREATE TABLE SALES(
Sales_No INT PRIMARY KEY,
Salesname VARCHAR(50),
Branch VARCHAR(50),
Salesamount INT,
DOB DATE
);
```

| | | | | |
|---|---|---|---|---|
| 4 | 13:38:04 | CREATE TABLE SALES( Sales_No INT PRIMARY... | 0 row(s) affected | 0.014 sec |

## 3 Create DEPARTMENTS Table

```
CREATE TABLE DEPARTMENTS(
dept_no INT PRIMARY KEY,
dept_name VARCHAR(50),
dept_location VARCHAR(50)
);
```

| | | | | |
|---|---|---|---|---|
| 5 | 13:38:52 | CREATE TABLE DEPARTMENTS( dept_no INT P... | 0 row(s) affected | 0.0082 sec |

## 4 Create EMPLOYEES Table

```
CREATE TABLE EMPLOYEES(
emp_id INT PRIMARY KEY,
emp_name VARCHAR(50),
emp_salary INT,
dept_no INT,
FOREIGN KEY (dept_no) REFERENCES DEPARTMENTS(dept_no)
);
```

| | | | | |
|---|---|---|---|---|
| 6 | 13:39:32 | CREATE TABLE EMPLOYEES( emp_id INT PRIM... | 0 row(s) affected | 0.016 sec |

# DML PART

# FROM TEACHER TABLE

## 1 Insert Five Records

```
INSERT INTO TEACHER VALUES
('Anita',1,'2015-06-12','Mathematics','Delhi',50000),
('Rohit',2,'2016-07-10','Commerce','Mumbai',45000),
('Neha',1,'2018-01-15','Mathematics','Delhi',48000),
('Amit',3,'2019-03-22','Science','Pune',47000),
('Kiran',2,'2014-11-05','Commerce','Mumbai',52000);
```

```
7    15:27:51    INSERT INTO TEACHER VALUES ('Anita',1,'2015...    5 row(s) affected Records: 5  Duplicates: 0  Warnings...    0.010 sec
```

## 2 25% Increment (Mathematics)

```
UPDATE TEACHER
SET Salary = Salary + (Salary * 0.25)
WHERE DeptName = 'Mathematics';
```

```
10    15:29:07    UPDATE TEACHER SET Salary = Salary + (Salar...    2 row(s) affected Rows matched: 2  Changed: 2  Warn...    0.0023 sec
```

## 3 15% Increment (Commerce)

```
UPDATE TEACHER
SET Salary = Salary + (Salary * 0.15)
WHERE DeptName = 'Commerce';
```

```
11    15:29:31    UPDATE TEACHER SET Salary = Salary + (Salar...    2 row(s) affected Rows matched: 2  Changed: 2  Warn...    0.0025 sec
```

# FROM SALES TABLE

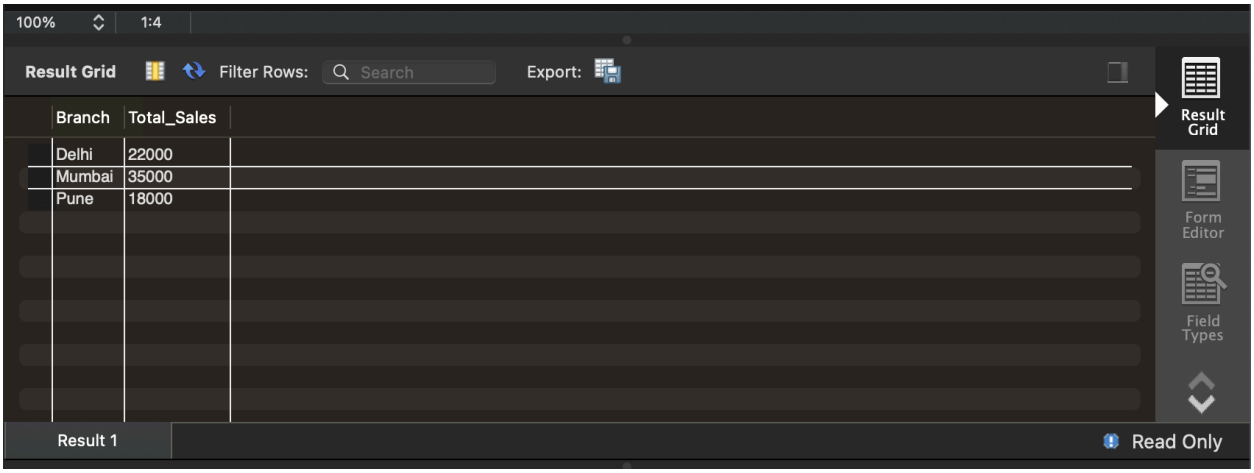
## 1 Insert Five Records

```
INSERT INTO SALES VALUES
(1,'Rahul','Delhi',10000,'2000-12-21'),
(2,'Amit','Mumbai',15000,'1998-05-10'),
(3,'Neha','Delhi',12000,'1999-12-11'),
(4,'Karan','Pune',18000,'1997-08-15'),
(5,'Simran','Mumbai',20000,'2001-12-05');
```

```
12    15:29:54    INSERT INTO SALES VALUES (1,'Rahul','Delhi',1...    5 row(s) affected Records: 5  Duplicates: 0  Warnings...    0.0034 sec
```

## 2 Total Sales Amount in Each Branch (GROUP BY)

```
SELECT Branch, SUM(Salesamount) AS Total_Sales
FROM SALES
GROUP BY Branch;
```



| Branch | Total_Sales |
|---|---|
| Delhi | 22000 |
| Mumbai | 35000 |
| Pune | 18000 |

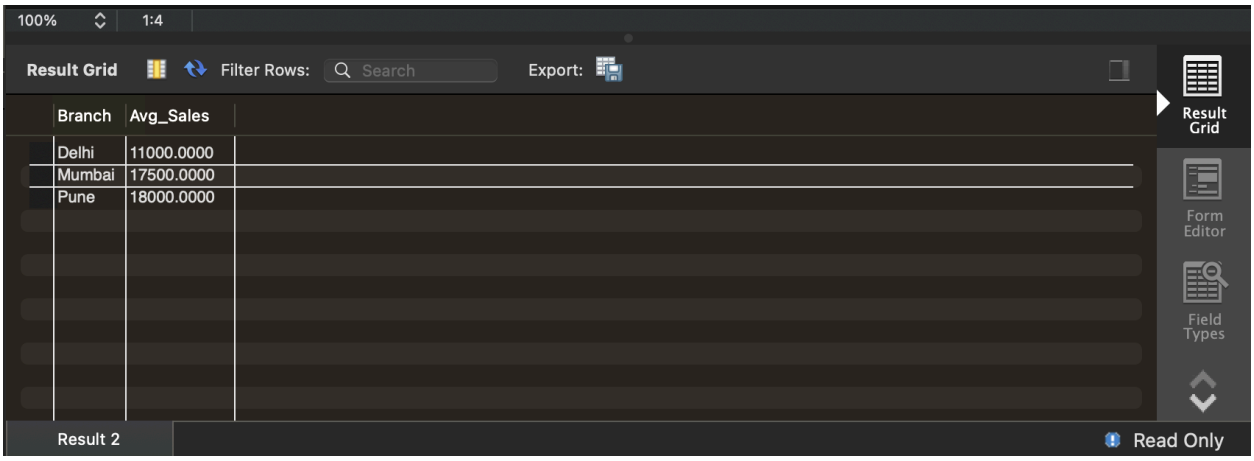
## 3 Average Sales Amount in Each Branch

```
SELECT Branch, AVG(Salesamount) AS Avg_Sales
FROM SALES
GROUP BY Branch;
```

---

## 4 December Born Salesmen (Formatted Date)

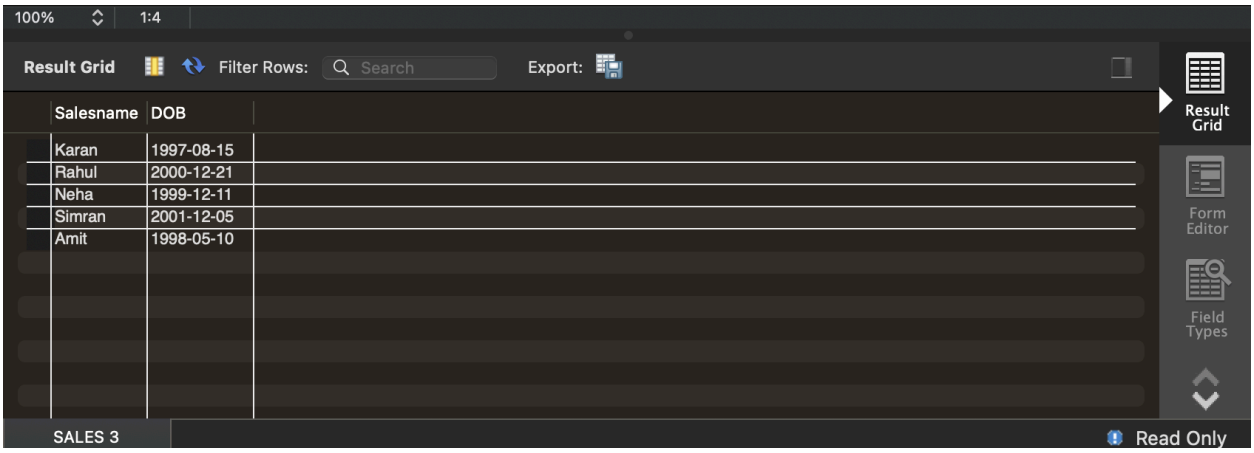
```
SELECT Salesname,
DATE_FORMAT(DOB,'%d-%b-%y') AS DOB
FROM SALES
WHERE MONTH(DOB)=12;
```



| Branch | Avg_Sales |
|---|---|
| Delhi | 11000.0000 |
| Mumbai | 17500.0000 |
| Pune | 18000.0000 |

## 5 Order By Month Alphabetically

```
SELECT Salesname, DOB
FROM SALES
ORDER BY MONTHNAME(DOB);
```



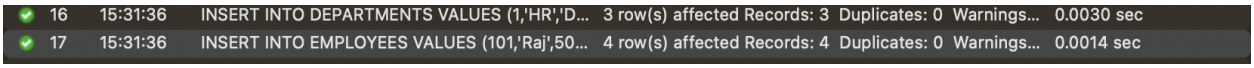
| Salesname | DOB |
|---|---|
| Karan | 1997-08-15 |
| Rahul | 2000-12-21 |
| Neha | 1999-12-11 |
| Simran | 2001-12-05 |
| Amit | 1998-05-10 |

# JOIN OPERATIONS

## Sample Data Insert (Departments & Employees)

```
INSERT INTO DEPARTMENTS VALUES
(1,'HR','Delhi'),
(2,'IT','Mumbai'),
(3,'Finance','Pune');

INSERT INTO EMPLOYEES VALUES
(101,'Raj',50000,1),
(102,'Priya',60000,2),
(103,'Arjun',55000,3),
(104,'Sneha',45000,2);
```

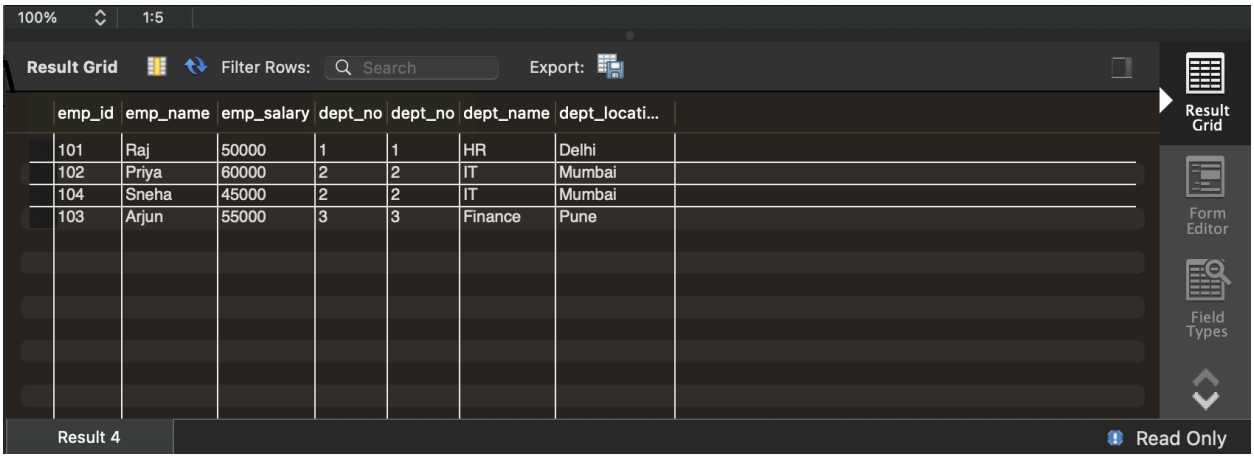


| | | | | | | |
|---|---|---|---|---|---|---|
| 16 | 15:31:36 | INSERT INTO DEPARTMENTS VALUES (1,'HR','D... | 3 row(s) affected Records: 3 | Duplicates: 0 | Warnings... | 0.0030 sec |
| 17 | 15:31:36 | INSERT INTO EMPLOYEES VALUES (101,'Raj',50... | 4 row(s) affected Records: 4 | Duplicates: 0 | Warnings... | 0.0014 sec |

## 1 INNER JOIN

```
SELECT EMPLOYEES.*, DEPARTMENTS.*
FROM EMPLOYEES
INNER JOIN DEPARTMENTS
```

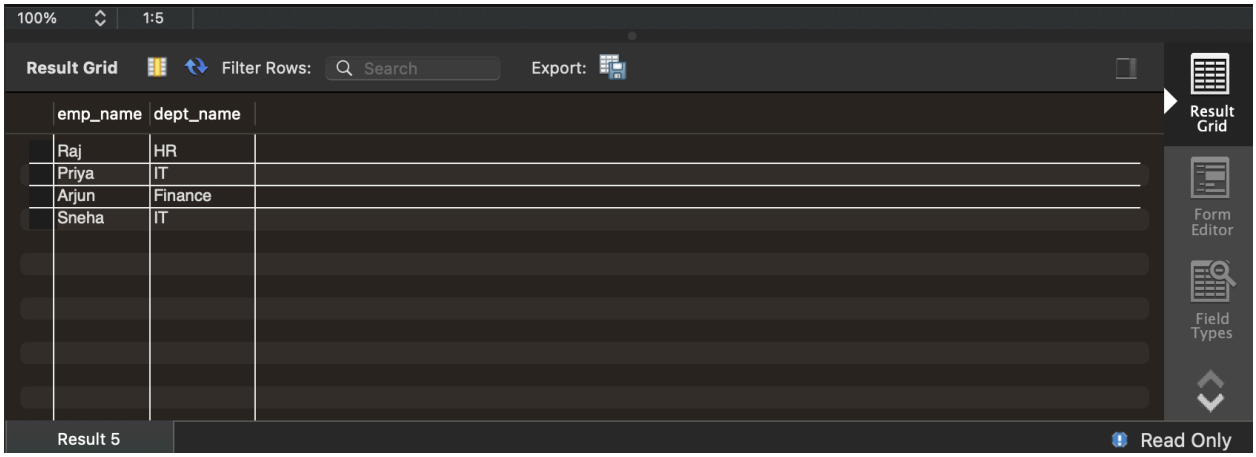
ON EMPLOYEES.dept_no = DEPARTMENTS.dept_no;



| emp_id | emp_name | emp_salary | dept_no | dept_no | dept_name | dept_locati... |
|---|---|---|---|---|---|---|
| 101 | Raj | 50000 | 1 | 1 | HR | Delhi |
| 102 | Priya | 60000 | 2 | 2 | IT | Mumbai |
| 104 | Sneha | 45000 | 2 | 2 | IT | Mumbai |
| 103 | Arjun | 55000 | 3 | 3 | Finance | Pune |

## 2 LEFT OUTER JOIN

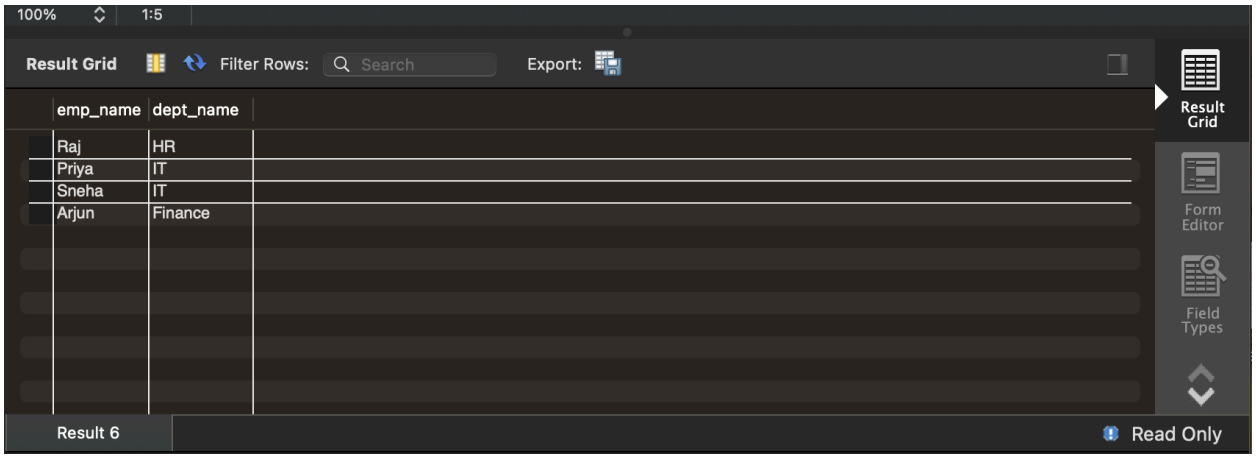
SELECT EMPLOYEES.emp_name, DEPARTMENTS.dept_name
FROM EMPLOYEES
LEFT JOIN DEPARTMENTS
ON EMPLOYEES.dept_no = DEPARTMENTS.dept_no;



| emp_name | dept_name |
|---|---|
| Raj | HR |
| Priya | IT |
| Arjun | Finance |
| Sneha | IT |

## 3 RIGHT OUTER JOIN

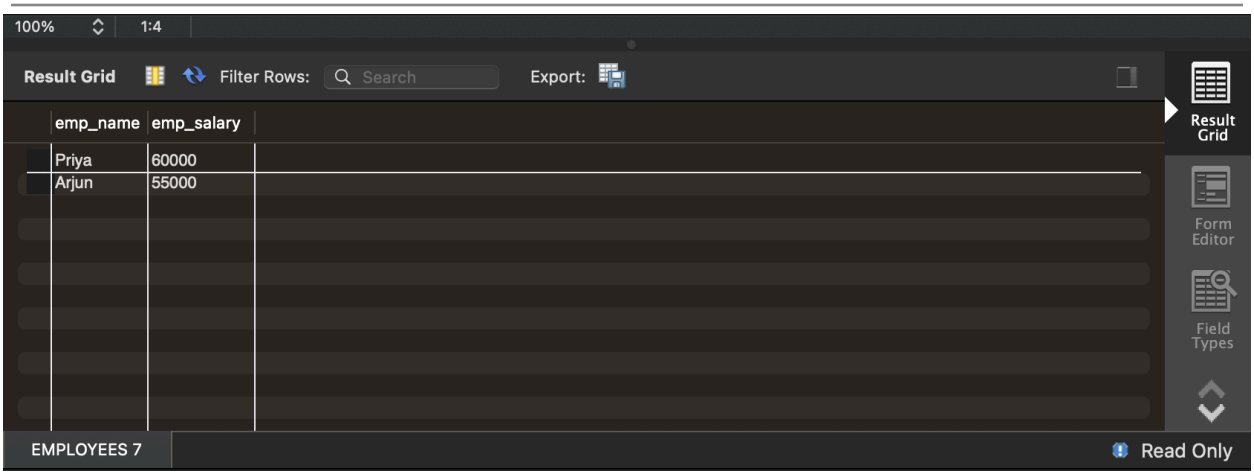
SELECT EMPLOYEES.emp_name, DEPARTMENTS.dept_name
FROM EMPLOYEES
RIGHT JOIN DEPARTMENTS
ON EMPLOYEES.dept_no = DEPARTMENTS.dept_no;



| emp_name | dept_name |
|---|---|
| Raj | HR |
| Priya | IT |
| Sneha | IT |
| Arjun | Finance |

## 4 Salary Greater Than Average

SELECT emp_name, emp_salary
FROM EMPLOYEES
WHERE emp_salary > (SELECT AVG(emp_salary) FROM EMPLOYEES);



| emp_name | emp_salary |
|---|---|
| Priya | 60000 |
| Arjun | 55000 |

# TCL (Transaction Control Language)

## 1 ROLLBACK

ROLLBACK;

## 2 COMMIT

COMMIT;

| | | | | |
|---|---|---|---|---|
| 23 | 15:33:46 | COMMIT | 0 row(s) affected | 0.00022 sec |

## 3 SAVEPOINT

SAVEPOINT A;

| | | | | |
|---|---|---|---|---|
| 24 | 15:34:15 | SAVEPOINT A | 0 row(s) affected | 0.00022 sec |

# DCL (Data Control Language)

## 1 Grant All Privileges

GRANT SELECT, INSERT
ON DBMS_Lakshita_Lab3.EMPLOYEES
TO 'labuser'@'localhost';

| | | | | |
|---|---|---|---|---|
| 28 | 15:35:40 | GRANT ALL PRIVILEGES ON DBMS_Lakshita_L... | 0 row(s) affected | 0.0031 sec |

## 2 Grant Selected Privileges

GRANT SELECT, INSERT
ON DBMS_Lakshita_Lab3.EMPLOYEES
TO 'labuser'@'localhost';

| | | | | |
|---|---|---|---|---|
| 31 | 15:36:37 | GRANT SELECT, INSERT ON DBMS_Lakshita_L... | 0 row(s) affected | 0.0017 sec |

## 3 Revoke All Privileges

REVOKE ALL PRIVILEGES
ON DBMS_Lakshita_Lab3.EMPLOYEES
FROM 'labuser'@'localhost';

| | | | | |
|---|---|---|---|---|
| 32 | 15:37:37 | REVOKE ALL PRIVILEGES ON DBMS_Lakshita_... | 0 row(s) affected | 0.0034 sec |

## 4 Revoke Selected Privileges

REVOKE INSERT
ON EMPLOYEES
FROM 'username'@'localhost';

| | | | | |
|---|---|---|---|---|
| 37 | 15:39:44 | GRANT INSERT ON DBMS_Lakshita_Lab3.EMP... | 0 row(s) affected | 0.0022 sec |

## CONCLUSION:

In this experiment, I learned how to use GROUP BY and ORDER BY clauses.I understood different JOIN operations.I practiced transaction control using COMMIT, ROLLBACK, and SAVEPOINT.I also learned how to manage privileges using GRANT and REVOKE commands.